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Sultanate of Oman

Ministry of Health

The Royal Hospital

Department of Child Health

Code: CH-DEV-PRT-2-Vers.1.1

Effective Date: 12/10/2023

Target Review Date: 10/10/2026

Title: Management of Spasticity using Botulinum Toxin Injection in Children

Statement:

This protocol is designed to be used as a guide to clinicians on the use of Botulinum toxin injections as part of management of spasticity within the pediatric population.

Spasticity:

Is a state of increased muscle tone that causes resistance to movement and impaired reflex function. It leads to stiffness, fibrosis and atrophy in muscles.

Spasticity has a disabling effect through pain and reduced movement and mobility, which can result in difficulties with walking, sitting, standing, and other everyday activities.

Botulinum Toxin:

Is a purified neurotoxin complex derived from the bacterium Clostridium botulinum. It stops muscle over activity by inhibiting the release of acetylcholine. It is given as an injection. It is broken into 7 neurotoxins (labeled as types A, B, C, D, E, F, and G), which are antigenically and serologically distinct but structurally similar. Human botulism is caused mainly by types A, B, E, and (rarely) F while types C and D cause toxicity only in animals.

Different products are available in the market.

- OnabotulinumtoxinA (Botox)
- AbobotulinumtoxinA (Dysport)
- IncobotulinumtoxinA (Xeomin)
- RimabotulinumtoxinB (Myobloc)

Indications of Toxin injection:

- Compromising care and hygiene
- Causing pain
- Impeding tolerance of other treatments, such as orthoses and use of equipment to support posture
- Causing cosmetic concerns to the child or young person.
- Disturbing sleep – Lower limb

Specific considerations:

- After an acquired non-progressive brain injury if rapid-onset spasticity is causing postural or functional difficulties.
- If focal dystonia is causing serious problems, such as postural or functional difficulties or pain.

Cautions:

- Bleeding disorder, for example due to anti-coagulant therapy
- Generalised spasticity
- Fixed muscle contractures
- Marked bony deformity or
- Lack of compliance with the post-treatment adapted physical therapy programme.

Contraindications:

- Has severe muscle weakness
- Had a previous adverse reaction or allergy to botulinum toxin type A
- Receiving aminoglycoside treatment.

Giving Information about:

- The possible benefits and the likelihood of achieving the treatment goals
- What the treatment entails, including:
 - * the need for assessments before and after the treatment
 - * the need to inject the drug into the affected muscles
 - * the possible need for repeat injections
 - * the benefits, where necessary, of analgesia, sedation or general anaesthesia
 - * the need to use serial casting or an orthosis after the treatment in some cases

Perform a careful assessment of muscle tone, range of movement and motor function to:

- Inform the decision as to whether treatment is appropriate
- Provide a baseline against which the response to treatment can be measured.
- A physiotherapist or an occupational therapist should be involved in the assessment.

- Topical or systemic analgesia or anaesthesia or sedation
- Consider ultrasound or electrical muscle stimulation to guide the injection of botulinum toxin type A.
- Ensure that maximum dosages are not exceeded.
- Ensure timely and follow up with physiotherapist and orthotics arranged.

Perform an assessment of muscle tone, range of movement and motor function:

- 6–12 weeks after injections to assess the response
- 12–26 weeks after injections to inform decisions about further injections.

Consider repeat injections of botulinum toxin type A if:

- Response was satisfactory, and the treatment effect has worn off.
- New goals amenable to this treatment are identified.

Dosing:

Botox:

- Maximum dose per patient and treatment session: 12U/kg b.w, but should not exceed 400U per patient.
- Dose range per muscle:
 - o Large muscle 3-6 U/Kg b.w.
 - o Small muscle 1-2U/Kg b.w.
 - o Should not exceed 50U per injection site or 100U per muscle.
- Re injection interval > 3 months

Dysport:

- Maximum dose per patient and treatment session: 15-30U/Kg b.w, should not exceed 1000U per patient.
- Dose range per muscle:
 - o Large muscle 6-18U/Kg b.w
 - o Small muscle 1-3U/kg b.w
- Re injection interval > 3 months

Treatment of Botulinum Toxin A toxicity:

- Most of the side effects if occurred are reversible. The most serious and feared side effect is generalized weakness (paralysis) which would require supportive treatment, depending on the degree of paralysis, patient might require non invasive or invasive ventilation.
- The role of Pyridostigmine in Botulinum toxin toxicity is still controversial.

STEP	PROCEDURE
1	Patients seen and assessed in the clinic. Information given to the patient/parents/family.
2	Patient booked for Botulinum toxin A injection as day care procedure.
3	On the day of Procedure; Patient advised to come fasting for 6 hours prior to procedure incase sedation needed.
4	Upon arrival to Daycare; patient will undergo routine nursing procedures including arthropometry and vitals check.
5	Clerking of the patient by the doctor excluding any recent illness or antibiotics use such as aminoglycoside. Written consent to be taken.
6	If fit for procedure, the doctor will proceed to localize the muscle groups using ultrasound guidance + application of local anesthetic agent.
7	If sedation is indicated: Oral midazolam for Child 1 month–18 years 500 micrograms/kg (max. 20 mg) 30–60 minutes before procedure to be given to the patient.(BNFC)
8	Preparation of the medication followed by injection of the toxin using ultrasound guidance.
9	If midazolam sedation is used: patient to be under monitor until fully awake and alert then can be discharged. If no midazolam used: patient can be discharged immediately after the injection.
10	Parents/Patient explained on what to look for at home and if any problems to report back to hospital. Follow up appointment with physiotherapy and the doctor arranged.

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
NICE guidelines: Spasticity in children and young adults with non-progressive brain injury. https://www.nice.org.uk/guidance/cg145/resources/spasticity-in-under-19s-management-35109572514757		2012	
Safety of high dose botulinum toxin type A therapy for treatment of pediatric spasticity. Edward M. Goldstein. Journal of child neurology 2006. http://www.rchsd.org/documents/2014/02/safety-of-high-dose-botulinum-toxin-type-a-therapy-for-the-treatment-of-pediatric-spasticity.pdf		2006	
National guidelines 2009: Spasticity in Adults: management using Botulinum toxin. http://shop.rcplondon.ac.uk/products/spasticity-in-adults-management-using-botulinum-toxin-national-guidelines		2009	

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Review Details

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